E.G.S.PILLAY ENGINEERING COLLEGE

(Autonomous)

Approved by AICTE, New Delhi | Affiliated to Anna University, Chennai

Accredited by NAAC with "A" Grade | Accredited by NBA (CSE, ECE,IT)

NAGAPATTINAM – 611 002

B.E. Biomedical Engineering

First Year – First Semester

Course Code	Course Name	L	T	P	C	Maximum Marks		
						CA	ES	Total
Theory Course								
1901MA104	Engineering Mathematics–I (Linear Algebra, Calculus and Partial differentiation)	3	1	0	4	40	60	100
1901CH103	Chemistry for Biomedical Engineering	3	0	0	3	40	60	100
1901GEX03	Programming for Problem Solving	3	0	0	3	40	60	100
1901ENX01	English for Engineers	2	0	0	2	100	-	100
Laboratory Course								
1901GEX52	Computer Programming Lab	0	0	2	1	50	50	100
1901GEX51	Engineering Intelligence I	0	0	2	1	50	50	100
1901CHX51	Engineering Chemistry Lab	0	0	2	1	50	50	100
1901HSX51	Communication Skills Lab	0	0	2	1	100	0	100
Total		**11**	**1**	**8**	**16**	**470**	**330**	**800**

L–Lecture |T–Tutorial| P–Practical| CA–Continuous Assessment |ES–End Semester

1901MA104	ENGINEERING MATHEMATICS–I (LINEAR ALGEBRA, CALCULUS AND PARTIAL DIFFERENTIATION) (ECE,MECH&BME)	L	T	P	C
		3	1	0	4

COURSE OBJECTIVES:

- To develop the use of matrix algebra techniques that is needed by engineers for practical applications.
- To familiarize the students with differential calculus.
- To familiarize the student with functions of several variables. This is needed in many branches of engineering.
- To make the students understand various techniques of integration.
- To acquaint the student with mathematical tools needed in evaluating multiple integrals and their applications.

COURSE OUTCOMES: After completion of the course, the student will be able to

CO1:	Calculate the nature of the matrix using Orthogonal Transformation.
CO2:	Develop the evolutes and envelopes of given curves by means of radius and centre of curvature.
CO3:	Calculate the area and volume of a curve using double and triple integration.
CO4:	Determine the nature of series using comparison, Ratio, Leibnitz tests.
CO5:	Examine the maxima/minima for the given function with several variables by finding stationary points

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	-	-	-	-	-	-	-	-
CO2	3	2	1	-	-	-	-	-	-	-	-	-
CO3	3	2	1	-	-	-	-	-	-	-	-	-
CO4	3	2	1	-	-	-	-	-	-	-	-	-
CO5	3	2	1	-	-	-	-	-	-	-	-	-
AVG	3	2	1	-	-	-	-	-	-	-	-	-

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	1	1	-
CO2	1	1	-
CO3	1	1	-
CO4	1	1	-
CO5	1	1	-
AVG	1	1	-

COURSE CONTENT

Module I	MATRICES	9 Hours

Inverse and rank of a matrix, rank-nullity theorem; System of linear equations;Symmetric,skew-symmetricandorthogonalmatrices;Determinants;Eigenvalues and Eigen vectors; Diagonalization of matrices; Cayley-Hamilton Theorem, and Orthogonal transformation

Module II	DIFFERENTIAL CALCULUS	9 Hours

Curvature in Cartesian co -ordinates – Centre and radius of curvature – Circle of curvature- Evolutes and involutes.

Module III	INTEGRAL CALCULUS	9 Hours

Double integration–Cartesian and polar coordinates–Change the order of Integration–Applications: Area of a curved surface using double integral–Triple integration in Cartesian co-ordinates – Volume as triple integral.

Module IV	SEQUENCES AND SERIES	9 Hours

Convergence of sequence and series, tests for convergence; Power series, Taylor's series, Series for exponential,

trigonometric and logarithm functions.		
Module V	**PARTIAL DIFFERENTIATION**	**9 Hours**
Partial derivatives, total derivative; Maxima, minima and saddle points; Method of Lagrange multipliers.		
Further readings:		
TEXTBOOKS:		

REFERENCES(BOOKS):

1. VeerarajanT. ,Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2018.
2. G.B.Thomas and R.L.Finney, Calculus and Analytic geometry,9th Edition, Pearson, Reprint, 2002.
3. Erwinkreyszig, Advanced EngineeringMathematics,9th Edition,JohnWiley&Sons,2006.
4. Ramana B.V., Higher Engineering Mathematics, Tata Mc Graw Hill New Delhi,11th Reprint, 2010.
5. D.Poole, Linear Algebra:A Modern Introduction,2nd Edition, Brooks/Cole, 2005.
6. N.P.Bali and Manish Goyal,A textbook of Engineering Mathematics, Laxmi Publications, Reprint, 2008.
7. B.S.Grewal, Higher Engineering Mathematics, Khanna Publishers, 36thEdition,2010.

1901CH103	CHEMISTRY FOR BIOMEDICAL ENGINEERING	L	T	P	C
		3	0	0	3

COURSE OBJECTIVES:

Imparting knowledge on the principles of nano technology, Drug delivery systems, materials for medicine, tissue engineering material and metals in biological system with their applications.

COURSE OUTCOMES: After completion of the course, the students will be able to

CO1:	Differentiate nano-particles, bulk materials, atoms and molecules
CO2:	Identify the suitable drug carriers for drug delivery
CO3:	Understand the quality of biomaterials as medicine
CO4:	Gain knowledge about alloys and polymers in tissue engineering
CO5:	Know the role of metals in biological components

COs Vs POs MAPPING:

COs	PO1	PO 2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	-	-	-	-	-	-	-	-	-
CO2	2	1	-	-	-	-	-	-	-	-	-	-
CO3	2	1	-	-	-	-	-	-	-	-	-	-
CO4	2	1	-	-	-	-	-	-	-	-	-	-
CO5	2	1	-	-	-	-	-	-	-	-	-	-
AVG	2	1	-	-	-	-	-	-	-	-	-	-

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	-	-
CO2	-	-	-
CO3	-	-	-
CO4	-	-	-
CO5	-	-	-
AVG	-	-	-

Module I	BASICS OFNANOCHEMISTRY	9 Hours

Nano technology Basics - distinction between molecules, nano particles and bulk materials; size-dependent properties.: nano cluster, nano rod, nano-tube(CNT)and nanowire. Synthesis: precipitation, thermolysis, hydrothermal, solvo thermal, electro deposition, chemical vapor deposition, laser ablation; Properties and applications.

Module II	DRUG DELIVERY SYSTEMS	9 Hours

Drug Delivery Systems -Some examples of drug carriers – cyclo dextrin- MOF- Pharmaceutical Co crystals. Fundamentals of Drug Nano particles: Delivery of Nano particles: Brain Delivery, Ocular Drug Delivery, Gene Delivery Systems, Carriers in Cancer Therapy, Cardiovascular System, Vascular Delivery to the Lungs, Targeting Lymphatics.

Module III	MATERIALS FOR MEDICINE	9 Hours

Materials for Medicine - General requirements of biomaterials- Polymer biomaterials, Elastomers, Thermoplastics, Thermosets, Applications of biomaterials - Cardiovascular devices - Orthopedic devices.

Module IV	MATERIALS FOR TISSUE ENGINEERING	9 Hours

Tissue engineering materials- Introduction to biomaterials -Natural materials-ECM(Collagen, elastin, glycoproteins, glycol saminoglycans, proteoglycans, Cross linked collagenpatch synthetic materials,-Expanded polytetrafluoro ethylene -Polyethylene terephthalate-Polyurethane. Organic inorganic Hybrid materials used in tissue engineering. Metals for tissue engineering-Titanium, stainless steel, nitinol, cobalt–chromium and platinum–chromium.

Module V		**9 Hours**
	Total	**45 Hours**

Further readings:

References:

1. Dara S.S,Umare S.S,―Engineering Chemistry‖, S.Chand & Company Ltd.,New Delhi 2010.
2. Sivasankar B.,―Engineering Chemistry ‖,Tata Mc Graw-Hill Publishing Company, Ltd., New delhi 2010
3. Gowariker V.R.,Viswanathan N.V.and JayadevSreedhar,―Polymer Science‖,NewAge

Ozin G. A. and Arsenault A. C., ―Nanochemistry: A Chemical Approach to Nanomaterials ‖, RSC Publishing,2005

Text Book:

1.Textbook of bio inorganic chemistry, R. K Sharma Delhi.

2.Modern Inorganic Chemistry By James E Huheey 2. Inorganic Chemistry By D F Shriver and P W Atkins1.

3.Nano-particle Technology for Drug Delivery. Edited by Ram B. Gupta, Uday B. Kompella, 2006, Taylor & Francis Group, 270 Madison Avenue, New York, NY 10016.

4.Tissue Engineering, Clemens van Blitterswijk, Peter Thomsen, Anders Lindahl, Jeffrey Hubbell, David Williams, Ranieri Cancedda, Joost de Bruijn, Jérôme Sohier,Academic Press, Elsevier, 84 Theobald’s Road, London WC1X 8RR, UK, 30 Corporate Drive, Suite 400,Burlington,MA01803,USA,525BStreet,Suite1900,SanDiego,CA92101-4495, USA,2008 ISBN:978-0-12-370869-4.

REFERENCES(WEBSITES):

1. https://www.ccdc.cam.ac.uk/solutions/csd-system/components/csd/
2. onlinelibrary.wiley.com/doi/10.1002/9780470661345.smc107/pdf

1901GEX03	PROGRAMMING FOR PROBLEM SOLVING (Common for all B.E./B.Tech Programme)	L	T	P	C
		3	0	0	3

COURSE OBJECTIVES:

1. To prepare students to comprehend the fundamental concepts
2. To demonstrate fine grained operations in number system
3. To gain exposure in programming language using C
4. To develop programming skills using the fundamentals and basics of C Language

FURTHER READING:

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1: Describe basic concepts of computers

CO2: Paraphrase the operations of number system CO3:Describe about basic concepts of C-Language

CO4:Understand the code reusability with the help of user defined functions

CO5:Analyze the structure concept, union, file management and preprocessor in C language

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	-	-	-	2	2	-	-	1
CO2	3	2	2	-	-	-	-	2	2	-	-	1
CO3	3	2	2	-	-	-	-	2	2	-	-	1
CO4	3	2	2	-	-	-	-	2	2	-	-	1
CO5	3	2	2	-	-	-	-	2	2	-	-	1
AVG	3	2	2	-	-	-	-	2	2	-	-	1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	1	-	1
CO2	1	-	1
CO3	1	-	1
CO4	1	-	1
CO5	1	-	1
AVG	1	-	1

Module I	INTODUCTION TO PROGRAMMING	9 Hours

Components of Computers andits Classifications- Problem Solving Techniques – Algorithm-Flowchart–Pseudo code – Program-Compilation -Execution

Module II	BASICS OF C PROGRAMMING	9 Hours

Structure of C program-C programming: Data Types–Storage classes-Constants –Enumeration Constants- Keywords – Operators: Precedence and Associativity - Expressions - Input/output statements – Decision making statements - Switch statement - Looping statements – Pre-processor directives.

Module III	ARRAYS AND STRINGS	9 Hours

Introduction to Arrays: Declaration, Initialization–One dimensional array–Two dimensional arrays– Example Program: Matrix Operations- String operations

Module IV	FUNCTIONS AND POINTERS	9 Hours

Introductiontofunctions:Functionprototype,functiondefinition,functioncall,Built-infunctions– Recursion – Example Program – Pointers – Pointer operators – Pointer arithmetic – Arrays and pointers – Array of pointers – Example Program: Sorting of names – Parameter passing: Pass by value, Pass by reference – Example Program: Swapping of two numbers and changing the value of a variable using pass by reference		
Module V	**STRUCTURES & FILE PROCESSING**	**9 Hours**
Structure-Nested structures–Pointer and Structures–Array of structures–Example Program using structures and pointers–Dynamic memory allocation-Files–Types-File processing: Sequential access, Random access-Command line arguments		
		Total:45Hours
Further readings: Object Oriented Programming Approach.		
References:		

1. Paul Deitel and Harvey Deitel,―C How to Program‖, Seventh edition, Pearson Publication
2. Juneja,B.Land Anita Seth,―Programming in C‖,CENGAGE Learning India pvt.Ltd.,2011
3. Pradip Dey,Manas Ghosh,―Fundamentals of Computing and Programming in C‖, First Edition,OxfordUniversityPress,2009.
4. Anita Goel and Ajay Mittal, ―Computer Fundamentals and Programming in C‖, Dorling Kindersley(India) Pvt. Ltd., Pearson Education in South Asia,2011.
5. Byron S.Gottfried," Schaum's Outline of Theory and Problems of Programming with C",Mc Graw-Hill Education,1996.

1901ENX01	ENGLISH FOR ENGINEERS (Common for all B.E./B.Tech. Programme)	L	T	P	C
		3	0	0	3

COURSE OBJECTIVES:

1.To develop the ability to read and comprehend technical texts in the field of Engineering
2.To develop vocabulary building through the study of word construction
3. To develop ability to write formal definitions of technical terms and expression.
4. To recognize various grammatical structures that will aid the student improve his/her theoretical knowledge.

COURSE OUTCOMES:

On the successful completion of the course, Students will be able to

CO1: Use appropriate words in a professional context
CO2: Gain understanding of basic grammatical structures and use them in right context.
CO3: Read and interpret information presented in tables, charts and other graphic forms.
CO4: Write definitions, descriptions, narrations and essays on various topics
CO5: Speak fluently and accurately in formal and informal communicative contexts.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	-	-	-	-	3	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-
AVG	-	-	-	-	-	-	-	-	-	3	-	-

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	-	-
CO2	-	-	-
CO3	-	-	-
CO4	-	-	-
CO5	-	-	-
AVG	-	-	-

Module I	FOCUS ON LANGUAGE (Vocabulary and Grammar)	9 Hours

Vocabulary -The Concept of Word Formation - Prefixes- Suffixes- Synonyms – Antonyms - Grammar - Articles-Preposition-Adjective-Adverb-Connectives-Tenses (present, past& future)-Conditional Clauses- Active voice –passive voice and Impersonal passive voice - Questions.

Module II	LISTENING SKILLS	9 Hours

Listening-Types of Listening -listening to short or longer texts- listening and Note taking- -formal and informal conversations- telephonic etiquettes- narratives from different sources. - Correlative verbal and nonverbal communication - listening to panel members (how to response to panel members after listening panel members) – listening to facing online interviews (or) interviews on video conferencing mode - listening webinars.

Module III	SPEAKING SKILL	9 Hours

Speaking - Stress and intonation –Communication skills-Role of ICT in Communication, -Process of communication- oral presentation skills-verbal and non verbal communication-individual and group presentations-imprompt presentation-publics peaking-Group discussion-speaking to the panel members (online interviews , video conferencing, online meeting and webinars.

Module IV	**READING SKILLS**	**9 Hours**
Reading–Intensive Reading –Predicting the content -Comprehending general and technical articles -Cloze reading-Inductivereading-Shortnarrativeanddescriptionsfromnewspapers–Skimmingandscanning-reading and interpretation-critical reading interpreting and transferring graphical information- sequencing of sentences- analytical reading on various Projects.		
Module V	**WRITING SKILLS**	**9 Hours**
Writing- Precise writing –Summarizing- Interpreting visual texts (pie chart, bar chart, picture, advertisements etc., -Proposalwriting(launchingnewunitsordepartmentinainstitutionorindustry& to get loan from bank) -Report writing (accident, progress, project, survey, Industrial visit)- job application- e-mail drafting- letter writing (permission, accepting and decaling)- e.mail drafting instructions – recommendations–checklist-uses of Print and electronic media (internet, fax, mobile, interactive video and teleconferencing, computer) e-governance.		
		Total:45Hours

REFERENCES:

1. Raman, Meenakshi and Sangeetha Sharma,―Technical Communication: Principles and Practice‖, Oxford University Press, New Delhi, 2011.
2. Rizvi and Ashraf M.,―Effective Technical Communication‖, Tata McGraw-Hill,NewDelhi,2005.
3. G. Radhakrishna Pillai,―English for Success‖, Central Institute of English and Foreign Languages‖, Emerald Publishers,Hyderabad,2003
4. Jones,D,―The Pronunciation of English‖,CUP,.Cambridge,2002.

1901GEX52	COMPUTER PROGRAMMING LAB (Common for all B.E./B.Tech. Programme)				
		0	0	2	1

COURSE OBJECTIVES:

1. To understand the constructs of C Language.
2. To develop C programmes using arrays and strings
3. To develop modular applications in C using functions and pointers
4. To develop applications in C using structures and union

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:Demonstrate the knowledge about the techniques used to solve problems in computing
CO2: Build programmes using C constructs.
CO3: Design and implement applications using arrays and strings
CO4: Develop and implement modular applications in C using functions and pointers.
CO5: Develop programmes and applications in C using structures, union.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	-	-	-	2	2	-	-	2
CO2	3	2	2	-	-	-	-	2	2	-	-	2
CO3	3	2	2	-	-	-	-	2	2	-	-	2
CO4	3	2	2	-	-	-	-	2	2	-	-	2
CO5	3	2	2	-	-	-	-	2	2	-	-	2
AVG	3	2	2	-	-	-	-	2	2	-	-	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	1	1	-
CO2	1	1	-
CO3	1	1	-
CO4	1	1	-
CO5	1	1	-
AVG	1	1	-

List of Experiments:

1.Working with word and style sheets
2.Write a C program to implement basic concepts
3.Write a C program to implement Decision Making and Branching statements
4.Write a C program to implement looping statements
5.Write a C program to implement Arrays
6.Write a C program to implement Strings
7.Write a C program to implement pointers
8.Write a C program to implement Structures
9.Write a C program to work with files in C

Total:45Hours

References

1.PaulDeitel and Harvey Deitel,―C How to Program‖, Seventh edition, Pearson Publication
2.Juneja,B.Land Anita Seth,―Programming in C‖,CENGAGELearningIndiapvt.Ltd.,2011
3.Pradip Dey,Manas Ghosh,―Fundamentals of Computing and Programming in C‖,First Edition, Oxford University Press, 2009.
4.AnitaGoelandAjayMittal,―Computer Fundamentals and Programming in C‖,Dorling Kindersley (India) Pvt. Ltd., Pearson Education in South Asia, 2011.
5.ByronS.Gottfried,"Schaum'sOutlineofTheoryandProblemsofProgrammingwithC",McGraw- Hill Education, 1996.

1901GE151	ENGINEERING INTELLIGENCE I (Common for all B.E./B.Tech. Programme)	L	T	P	C
		0	0	2	1

COURSE OBJECTIVES:

To equip students with the knowledge and skills to design, develop, and deploy intelligent systems

Course Outcomes:

At the end of the course, students will be able to

CO1:Apply their knowledge and skill to engineering field

CO2: Understand the value of individual competence

CO3: Apply their skill to career planning and team work

CO4: Illustrate verbal and non verbal skills

CO5:Use various communication skill exercise to write and interpret the contents

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	2	2	-	-	-	2	2
CO2	3	3	2	-	-	2	2	-	-	-	2	2
CO3	3	3	2	-	-	2	2	-	-	-	2	2
CO4	3	3	2	-	-	2	2	-	-	-	2	2
CO5	3	3	2	-	-	2	2	-	-	-	2	2
AVG	3	3	2	-	-	2	2	-	-	-	2	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	2	-
CO2	-	2	-
CO3	-	2	-
CO4	-	2	-
CO5	-	2	-
AVG	-	2	-

Module I	**BEHAVIORALCHANGES–TRANSITIONOFSCHOOL TO COLLEGE**	**6 Hours**

Vocabulary -The Concept of Word Formation - prefixes- suffixes- Synonyms – Antonyms - Grammar - Articles-Preposition-Adjective-Adverb-connectives-Tenses (present, past & future)-Sentence pattern-Types of sentences –Active voice–passive voice and Impersonal passive voice-Wh-Questions.

Module II	**EXPOSURETO INDIVIDUALCOMPETANCE**	**6 Hours**

Listening- listening intently-arousing and sustaining interest-listening to short or longer texts- formal and informal conversations-telephonic etiquettes –narratives from different sources.-listening and Note taking-correlative verbal and non verbal communication-listening to TOEFL&IELTS programs-listening to Project presentation-listening to technical seminar and conferences.

Module III	**CAREERPLANNING**	**6 Hours**

Speaking - stress and intonation –persuasive speaking -Describing person, place and thing - sharing personal information — greetings –taking leave -Individual and Group Presentation-impromptu Presentation-public speaking-Group Discussion-project planning-facing viva voce and delivering project.

Module IV	**INTRODUCTION TOCOMMUNICATION SKILLS**	**6 Hours**

Reading– comprehending general and technical articles -cloze reading -inductive reading- short narrative and descriptions from newspapers – Skimming and scanning-reading and interpretation-critical reading interpreting and transferring graphical information-sequencing of sentences-analytical reading on various Projects.

Module V	**COMMUNICATIONEXERCISE-1**	**6 Hours**

Writing- Precise writing –Summarizing- interpreting visual texts (pie chart, bar chart, picture – advertisements etc., -Proposal writing(launching new units or department in a institution or industry & to get loan from bank) -report writing (accident, progress, project, survey, Industrial visit)- job application- e-mail drafting- letter writing (permission, accepting and decaling)-instructions –recommendations– checklist
Total:30Hours
References:
1.Dr.P.Prasad (2012)―The Functional Aspects of COMMUNICATION SKILLS‖; fifth Edition; S.K Kataria & Sons Publication
2.Kalyana;(2015)―Soft Skill for Managers ‖; First Edition; Wiley Publishing Ltd.
3.Aruna Koneru (2008)―Professional Communication‖;Second edition;Tata Mc Graw-Hill Publishing Ltd.

1901CHX51	ENGINEERING CHEMISTRY LAB	L	T	P	C
		0	0	2	1

COURSE OBJECTIVES:

Engineering Chemistry laboratory course is designed to provide basic chemistry and its application to the first year engineering students. The course includes the study of applications of water quality chemistry, identification of acidic and alkaline nature of water, molecular weight determination and explaining the principles behind each experiments.

List of Practical Experiments

1. Determination of total, temporary & permanent hardness of water by EDTA method
2. Determination of strength of given hydrochloric acid using pH meter
3. Estimation of iron content of the given solution using potentiometer
4. Estimation of sodium present in water using flame photometer
5. Corrosion experiment–weight loss method
6. Determination of molecular weight of a polymer by visco metry method
7. Conducto metric titration of strong acid Vs strong Base
8. Estimation of dissolved oxygen in a water sample/sewage by Winklers method.
9. Comparison of alkalinities of the given water samples
10. Determination of concentration of unknown colored solution using spectro photometer
11. Determination of percentage of copper in alloy
12. Determination of ferrous iron in cement by Spectro photometry method
13. Adsorption of acetic acid on charcoal
14. Determination the flash point and fire point of a given oil using Pensky martine closed cup apparatus
15. Determination the calorific value of solid fuels
16. Determination the structural of the compound using chemo software.

COURSE OUTCOMES

After completion of the course, the student will be able to

CO1:Measure the hardness and alkalinity of given water sample

CO2:Find the amount and percentage of iron in unknown sample using EMF and photometric methods

CO3:Determine the amount of strong acid present in the given sample using PH metric and conducto metric methods

CO4:Determine the amount of dissolved oxygen and heavy metal present in the given sample

CO5: Determine the molecular weight of the given polymer

COs Vs POs MAPPING:

COs	PO1	PO 2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		-	1	-	-	3	3	-	-	-
CO2	3	2		-	1	-	-	3	3	-	-	-
CO3	3	2		-	1	-	-	3	3	-	-	-
CO4	3	2		-	1	-	-	3	3	-	-	-
CO5	3	2		-	1	-	-	3	3	-	-	-
AVG	3	2		-	1	-	-	3	3	-	-	-

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	1	-	1
CO2	1	-	1
CO3	1	-	1
CO4	1	-	1
CO5	1	-	1

AVG	1	-	1

TEXTBOOKS:

1. Experimental organic chemistry,Daniel R.Palleros, John Wiley&Sons, Inc.,NewYor(2001)
2. ―Engineering Chemistry‖,Jain&Jain,15th edition,Dhanpat Rai Publishing company,New Delhi.
3. Vogel"s Text book of practical organic chemistry, Furniss B.S.Hanna ford A.J,Smith P.W.Gand Tatchel A.R LBS Singapore (1994).
4. LBS Singapore(1994).KolthoffI.M., Sandell E.B.etal Mcmillan,Madras1980.

1901HSX51	COMMUNICATION SKILLS LAB (Common to all B.E./B.Tech. Programme)	L	T	P	C
		0	0	2	1

COURSE OBJECTIVES:
This Lab focuses on using multi-media instruction for language development to meet the following targets:

1. To improve the students‟ fluency in English, through a well-developed vocabulary and enable them to listen to English spoken at normal conversational speed by educated English speakers and respond appropriately in different socio-cultural and professional contexts
2. Further, they would be required to communicate their ideas relevantly and coherently in writing.
3. To prepare all the students for their placements

COURSE OUTCOMES:
On the successful completion of the course, students will be able to
CO1: Improve their listening, reading, speaking and writing skills.
CO2: Develop their communication competency.
CO3: Use language effectively in professional contexts.
CO4: Develop the ability to face campus interviews.
CO5: Use language efficiently in expressing their opinions

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	-	-	-	-	3	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-
AVG	-	-	-	-	-	-	-	-	-	3	-	-

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	-	-
CO2	-	-	-
CO3	-	-	-
CO4	-	-	-
CO5	-	-	-
AVG	-	-	-

List of Experiments:

1. **Activities on Fundamentals of Listening and Inter-personal Communication(6)**
   Listening to conversation, listening to technical presentation- listening to online video conferencing ,interviews and webinars-starting a conversation - responding appropriately and relevantly - using appropriate body language - Role Play in different situations & Discourse Skills- using visuals.
2. **Activities on Reading Comprehension** **(6)**
   General Vs Local comprehension- reading for facts- guessing meanings from context-Scanning-skimmingandinferringmeaning-criticalreading&effectivegoogling- TOFEL,IELTS-reading online journals.
3. **Activities on Writing Skills** **(6)**

Structure and presentation of different types of writing - letter writing - Resume writing-e- correspondence - Proposal writing - Technical report writing - Portfolio writing - planning for writing - improving one's writing.

4. Activities on Presentation Skills **(6)**

Oral presentations (individual and group) through JAM sessions – presentation on online platform (webinars, online meeting) - seminars -PPTs and written presentations through posters- projects- report- e-mails- assignments etc.- creative and critical thinking.

5. Activities on Soft Skills **(6)**

Dynamics of group discussion, intervention, summarizing, modulation of voice, body language, relevance, fluency and organization of ideas and rubrics for evaluation-Concept and process, pre- interview planning, opening strategies, answering strategies, interview through tele-conference & video-conferencing and Mock Interviews-Time management-stress management –paralinguistic features- Multiple intelligences – emotional intelligence – spiritual quotient (ethics) – intercultural communication – creative and critical.

Total: 30Hours

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5. Management Shapers Series‖,Universities Press (India)PvtLtd., Himayat nagar, Hyderabad 2008.
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7. Jones,D,―The Pronunciation of English‖,CUP,.Cambridge,2002.